

Kill–Resist–Renew Trinity: Hyperbranched Polymer with Self-Regenerating Attack and Defense for Antifouling Coatings

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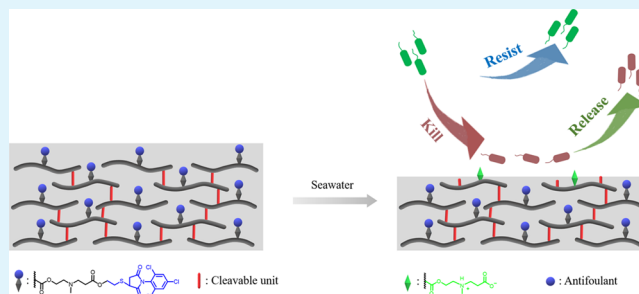
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ABSTRACT: Traditional antifouling coatings are generally based on a single antifouling mechanism, which can hardly meet the needs of different occasions. Here, a single “kill–resist–renew trinity” polymeric coating integrating fouling killing, resistance, and releasing functions is reported. To achieve the design, a novel monomer-tertiary carboxybetaine ester acrylate with the antifouling group *N*-(2,4,6-trichlorophenyl)maleimide (TCB-TCPM) is synthesized and copolymerized with methacrylic anhydride via reversible addition-fragmentation chain transfer polymerization yielding a degradable hyperbranched polymer. Such a polymer at the surface/seawater is able to hydrolyze and degrade to short segments forming a dynamic surface (releasing). The hydrolysis of TCB-TCPM generates the antifouling groups TCPM (killing) and zwitterionic groups (resistance). Such a polymeric coating exhibits a controllable degradation rate, which increases with the degrees of branching. The antibacterial assay demonstrates that the antifouling ability arise from the synergistic effect of “attacking” and “defending”. This study provides a new strategy to solve the challenging problem of marine biofouling.

KEYWORDS: antifouling coating, zwitterion, hyperbranched polymer, degradation, fouling resistance



INTRODUCTION

Marine biofouling, an undesirable colonization of marine organisms, has negative impacts on marine industries and marine activities.^{1–3} Two strategies are generally employed to solve the problem: one involves killing the biofouling close to the surface (attacking), and the other involves resisting adhesion of the biofouling to the surface (defending).^{4–6} Polymers containing antifouling groups such as quaternary ammonium salts,^{7–9} phosphonium salts,^{10–12} triclosan,^{13–15} isothiazolinone,^{16,17} and *N*-(2,4,6-trichlorophenyl) maleimide^{18,19} have been used to prepare “attacking” coatings. The surface of these “attacking” coatings exhibits efficient bactericidal properties via release–killing or contact–killing. However, the accumulation of dead bacteria or proteins on the surfaces would shield the antifouling moieties and yield a considerable decrease in their antifouling efficiency.^{6,15,20,21} On the other hand, zwitterionic polymers and poly(ethylene glycol)-based polymers are typical “defending” materials capable of preventing protein adsorption or initial bacteria adhesion for a certain period.^{22–27} These polymers cannot directly kill or inhibit the growth of bacteria when the bacteria attach to surfaces, and even a few attached bacteria will ultimately grow into a biofilm.^{28,29} Thus, surfaces based on an integrated “attacking and defending” strategy is proposed to improve the antifouling ability and prolong the duration of the polymers. Jiang et al. grafted switchable polymer brushes between “attacking” cationic *N,N*-dimethyl-2-morpholinone

and “defending” zwitterionic carboxy betaine from gold substrates.³⁰ The “attacking” surface can kill over 99% of bacteria *E. coli* K12 on the surface under dry conditions. The “attacking” surface can rapidly transform to a “defending” zwitterionic surface in neutral or alkaline aqueous environments, which can release dead bacteria and resist bacterial adhesion. Diblock polymer brushes consisting of zwitterionic and quaternary ammonium salt polymers were grafted on commercial membranes, and the surface can reduce the adhesion of protein and bacteria and decrease the cell viability of *E. coli* from 73 to 28%.³¹ However, such a grafting procedure is complex. Also, the zwitterionic polymers are highly hydrophilic and easily swell during immersion in marine environments, which declines their mechanical strength.³² Thus, antifouling coating with hydrophobic matrix is more suitable and persistent for the marine industry. Particularly, the “attacking” or “defending” surface is susceptible to inorganics or dead microorganisms, which could cover the surfaces in complex marine environments, leading to failure of the antifouling function.^{17,33,34} Accordingly, regeneration of the

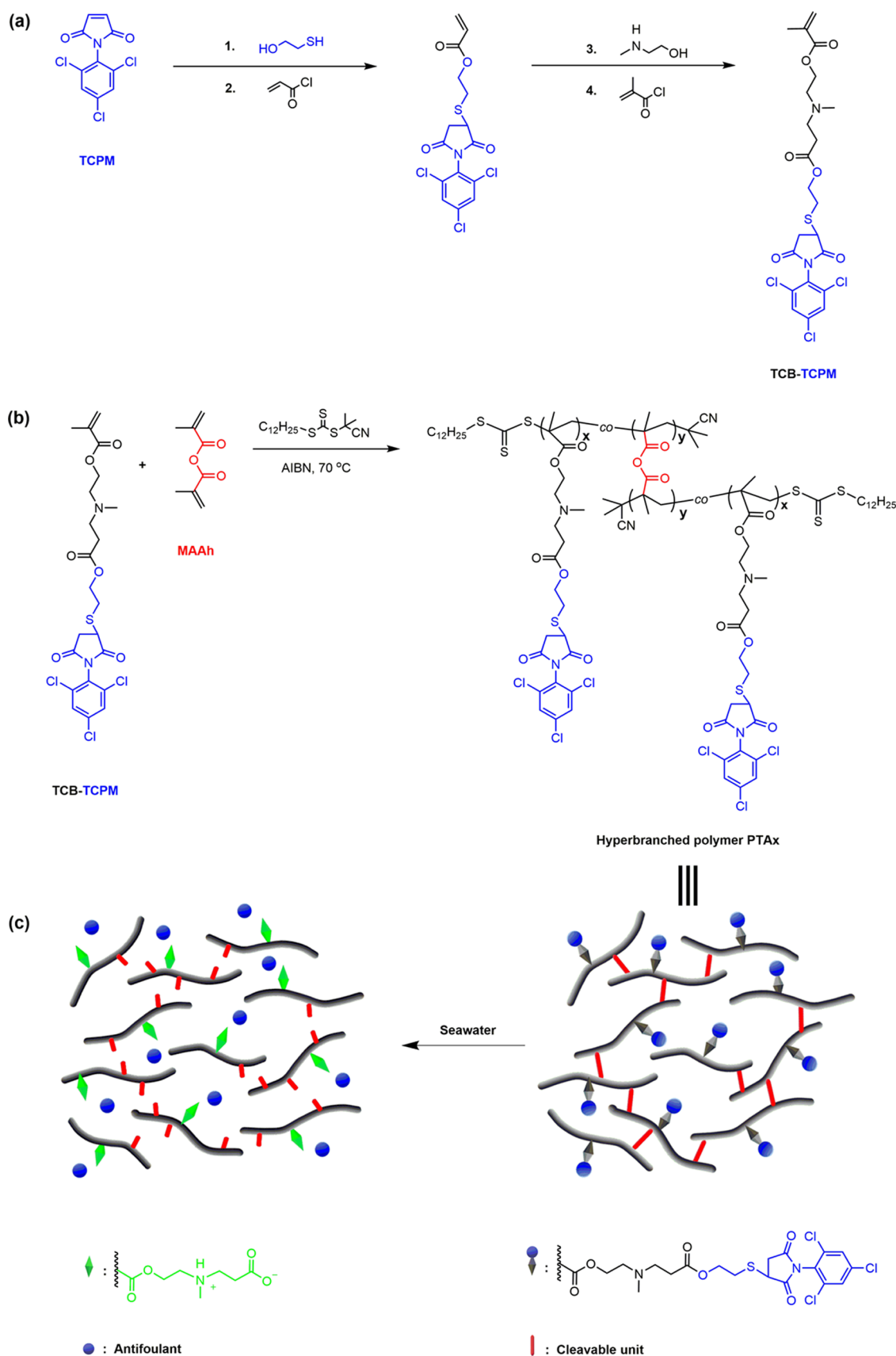
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Scheme 1. Synthesis of (a) TCB-TCPM and (b) PTAx and (c) Hydrolysis and Degradation of PTAx



antifouling surface after contamination is critical for improving its durability in marine environments. Many efforts have been

paid to designing a smart surface based on a “kill and release” strategy, which enables switchable transitions in the surface

properties by an external stimulus (e.g., pH, temperature, or light) to release the adhered microorganisms.^{21,31,35} However, it is difficult to provide the external stimulus in marine environments. Therefore, a spontaneous “release” function via hydrolysis or degradation has been proposed.^{33,36} It can form a changing surface and release the adhered bacteria, which is important for marine antifouling.

Self-polishing copolymers (SPCs) and degradable polymers are two types of polymers that can form a self-renewable or dynamic surface. For SPCs, their pendant groups can be hydrolyzed, thereby generating a hydrophilic surface, which can be polished away with the help of water flow.^{37,38} For degradable polymers, the main chains can degrade to oligomers or small molecules, which easily dissolve into seawater.^{39–41} The surface renewal of degradable polymers occurs spontaneously without an external shear force, leading to a better antifouling ability than SPCs during idle periods.⁴² Moreover, the surface continuously renews itself and can release the adhered organics or inorganics.^{42,43} However, both types of polymers exhibit a limited antifouling ability owing to the lack of fouling-resistant moieties. Generally, the addition of antifoulants is required for improving the antifouling ability, which may be harmful to the environment. Incorporation of “attacking and defending” moieties into polymers with the release function or self-renewal ability therefore represents a promising strategy for realizing durable marine antifouling.

So far, most antifouling polymers are linear and considerable solvents are required for preparing coatings with low viscosity, which would generate highly volatile organic compounds. Hyperbranched polymers with lower chain entanglement have higher solubility and more end-up functionalities relative to their linear analogs.^{44,45} Hyperbranched polymers as an antifouling coating probably have high performance. Divinyl monomers are often used as the branching agents since most of these agents are commercially available.^{46,47} In particular, for cleavable divinyl monomers, the branched architectures can fragment into short polymer chains, thereby yielding surface-fragmenting hyperbranched polymers that exhibit a controlled surface renewal ability.

In this work, we have synthesized a hydrophobic dual-functional monomer—tertiary carboxybetaine ester acrylate with an antifouling group *N*-(2,4,6-trichlorophenyl)maleimide (TCB-TCPM). This acrylate was then copolymerized with divinyl monomer—methacrylic anhydride to prepare degradable hyperbranched polymers via reversible addition–fragmentation chain transfer polymerization (RAFT). The hydrolysis of TCB-TCPM generates the antifouling groups TCPM to kill the biofouling (attacking) and zwitterionic groups to resist the adhesion of biofouling (defending). The transition from “attacking” to “defending” occurs on the surface, which maintains a hydrophobic matrix. Thus, the coating would maintain its good strength after immersion in seawater. Moreover, the copolymer at the surface/seawater can hydrolyze and degrade into small pieces, leading to a self-regenerating or dynamic surface, which can release the adhered organics or inorganics and regenerate the “attacking” and “defending” functional surface.⁴⁸ To our knowledge, it is the first reported hydrophobic polymer with self-regenerating “attacking” and “defending” functions for antifouling coating. We examine the coating properties, degradation behaviors, surface morphology, and anti-biofouling ability. Our aim is to develop a highly effective marine anti-biofouling system.

RESULTS AND DISCUSSION

TCB-TCPM was synthesized following the route shown in Scheme 1 (the corresponding ¹H NMR spectrum of the product is shown in Figure 1a). The typical peaks at 7.45 ppm,

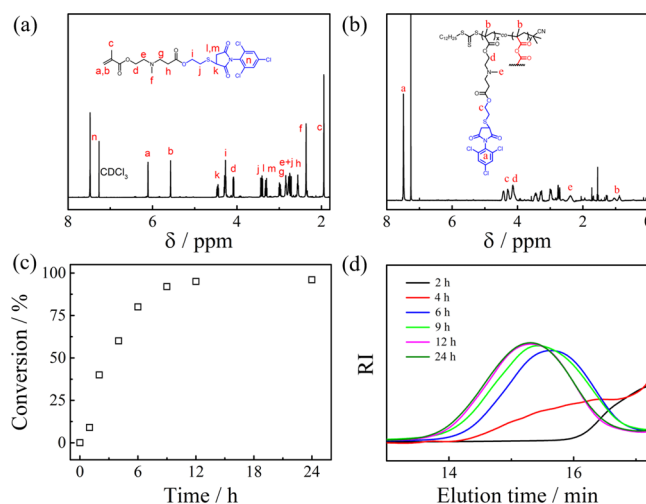


Figure 1. (a) ¹H NMR spectrum of TCB-TCPM; (b) ¹H NMR spectrum of PTAS; (c) time dependence of monomer conversion; (d) GPC elution trace of PTAS during the polymerization of PTAS.

the peaks at 2.30 ppm, and the peaks at 6.07 and 5.53 ppm are attributed to the phenyl group, methylamino group, and the vinyl group, respectively, and intensity ratio of the three typical peaks is 2:3:2, indicating the successful synthesis of TCB-TCPM. After this synthesis, we prepared hyperbranched polymers (PTAx) of TCB-TCPM and methacrylic anhydride (MAAh) via RAFT polymerization, where MAAh serves as the branched agent (*x*: feed molar percentage of MAAh). Figure 1b shows the ¹H NMR spectrum of PTAS. Typically, the peaks at 5.50–6.10 ppm attributed to the vinyl group disappeared and the peaks at 7.50–7.40 ppm attributed to the phenyl group in TCB-TCPM are observed. Peaks at 1.15–0.80 ppm attributed to the CH₃ in TCB-TCPM and MAAh can be observed, indicating that the hyperbranched copolymers of PTA_x have been successfully synthesized.

Note that MAAh can be hydrolyzed in aqueous environments, and hence the hyperbranched polymers would be degraded into oligomers, leading to surface fragmentation. To adjust the degradation rate of the polymers, different contents (ranging from 0 to 10%) of MAAh relative to all monomers are fixed. For comparison, 5% of the uncleavable divinyl monomer—ethylene glycol dimethacrylate (EGDMA) is used to prepare the control sample. We calculate the degrees of branching (DB) for PTA_x from the relationship given in the following equation:

$$DB = B / (B + L) \times 100\%$$

where *B* is the number of branching units and *L* is the number of linear units in the branched polymer. The DB value is calculated from the area of peaks *a* and *b* that occur in the ¹H NMR spectrum (Figure 1b): peak *a* (7.50–7.40 ppm) is attributed to the phenyl group of TCB-TCPM corresponding to *L* and peak *b* (1.15–0.80 ppm) is attributed to the methyl group of TCB-TCPM and MAAh corresponding to (*B* + *L*).

Table 1 shows that when [MAAh] increases from 2.5 to 10, the DB of PTA_x gradually increases from 1 to 12 mol %, which

Table 1. Characterization Data of the Polymers

sample	feed molar ratio ^a	cross-linker	reaction time (h)	polymer composition ^b	$M_n^c \times 10^{-3}$ (g/mol)	$M_w^c \times 10^{-3}$ (g/mol)	D_M^c	DB ^d (mol %)
PTA0	100/0/1/0.2	MAAh	24	100/0	20	34	1.7	0
PTA2.5	97.5/2.5/1/0.2	MAAh	24	99/1	21	33	1.6	1
PTA5	95/5/1/0.2	MAAh	24	94/6	19	30	1.6	6
PTA7.5	92.5/7.5/1/0.2	MAAh	24	91/9	20	34	1.7	9
PTA10	90/10/1/0.2	MAAh	24	88/12	19	32	1.7	12
PTE5	95/5/1/0.2	EGDMA	24	94/6	21	33	1.6	6
PTA5-6h	95/5/1/0.2	MAAh	6	94/6	15	23	1.5	6
PTA5-12h	95/5/1/0.2	MAAh	12	94/6	18	32	1.8	6

^aFeed molar ratio of TCB-TCPM/cross-linker/CPDT/AIBN. ^bPolymer molar composition of TCB-TCPM/cross-linker, as determined via ¹H NMR. ^cDetermined via GPC. ^dDegrees of branching (DB) were calculated from: $DB = B/(B + L)$, where B is the number of branching units and L is the number of linear units in the branched polymer, as determined by means of ¹H NMR.

indicates the presence of branched structures in the copolymers. We also examined the effects of the reaction time on the molecular weight and the DB. When the reaction time increases from 6 to 24 h, the M_n of the precipitated polymer increases from 1.5 to 1.9×10^4 g/mol and the DB remains at 6 mol %, which indicates the controlled growth of the hyperbranched polymers. We also monitored the polymerization kinetics and the time dependence of the molecular weight (Figure 1c,d). At 6 h, the polymerization follows approximately first-order kinetics. The GPC traces exhibit a gradual shift to lower elution time, indicating the molecular weight of the polymers increases with polymerization time. After 6 h, the polymerization rate slows down and the monomer conversion reaches $\sim 100\%$ at 24 h because of the gradually decreased monomer concentration. In addition, the GPC traces of the crude polymers show that the molar mass distribution is unimodal and narrow after 6 h, which indicates the polymerization was well controlled by the RAFT process.

We examined the degradation of the polymer coatings in artificial seawater (ASW). The sample was cast-coated on an epoxy panel, where the epoxy substrate has a slight effect on the coating property due to the sufficient thickness of the coating. As shown in Figure 2a, each coating starts to lose

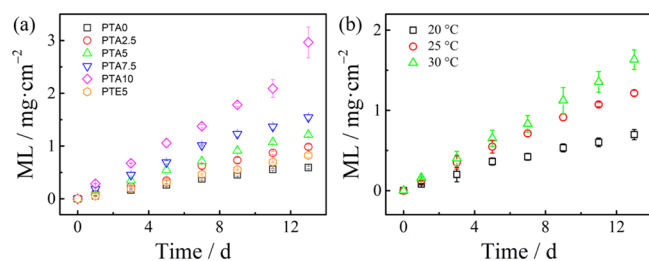


Figure 2. Time dependence of (a) PTAx and PTE5 mass loss (ML) in ASW at 25 °C and (b) PTA5 mass loss in ASW at different temperatures. Three parallel samples were measured to get the average value.

weight after immersion in ASW and the weight loss increases with increasing immersion time. For linear polymer PTA0, mass loss levels of ~ 0.16 , 0.37 , and 0.60 $\text{mg}\cdot\text{cm}^{-2}$, corresponding to an approximately linear erosion, occur after 3, 7, and 14 d, respectively. No cleavable points occur in the main chain of the coatings, and hence, the mass loss is attributed to the hydrolysis of TCB-TCPM. For hyperbranched polymer PTE5 with no cleavable branched points, mass loss values of ~ 0.19 , 0.49 , and 0.82 $\text{mg}\cdot\text{cm}^{-2}$ are obtained after 3, 7, and 14 d of immersion, respectively. In contrast,

these times yield mass loss levels of ~ 0.35 , 0.71 , and 1.20 $\text{mg}\cdot\text{cm}^{-2}$ for the hyperbranched polymer PTA5 with cleavable branched units. The larger mass loss of PTA5 compared with that of PTE5 indicates that the cleavage of the branching units makes the hyperbranched polymers degrade into short segments and easily dissolve into ASW, leading to a surface renewal. Moreover, the degradation of PTAx undergoes approximately linear mass loss during immersion in ASW, and this loss increases with increasing MAAH content. For PTA2.5, PTA5, PTA7.5, and PTA10, mass loss levels of ~ 0.99 , 1.20 , 1.54 , and 2.96 $\text{mg}\cdot\text{cm}^{-2}$, respectively, are gradually reached after 14 d of immersion in ASW. These results indicate that the degradability of PTAx can be tuned by the content of cleavable branched units. We also examined the changes of thickness for the PTAx in ASW at 25 °C (Figure S1). Clearly, the changes of thickness have a similar trend to that of mass loss, which indicates the polymeric coatings linearly erode during immersion in ASW. Considering that the temperature of marine environment changes with the season, we examined the mass loss of PTA5 in ASW under 20–30 °C (Figure 2b). The mass loss still increases linearly with the immersion time at different temperatures, indicating the coating cannot wholly collapse during the immersion time. After 14 d, the mass loss of PTA5 under 20, 25, and 30 °C is ~ 0.70 , 1.20 , and 1.65 $\text{mg}\cdot\text{cm}^{-2}$, respectively. This fact indicates that the degradation rate of the PTA coating increases with temperature. In other words, the coating exhibits a temperature-responsive ability. We also examined the release of TCPM during immersion in ASW at 25 °C (Figure S2). The release rate of TCPM remains at ~ 11 $\mu\text{g}\cdot\text{cm}^{-2}\cdot\text{d}^{-1}$, which indicates a constant release rate of the antifoulant. The cumulative release of TCPM is linearly related to immersion time, indicating the release of TCPM is in the first-order kinetics. Due to the linear degradation of the polymeric coating and the linear release of TCPM, the long-term anti-biofouling depends on the degradation rate and thickness of the coating. Theoretically, 200 μm of the PTAx coating in the present study can last for over 2.5 years according to the decrease of thickness (~ 2 to 6 $\mu\text{m}/\text{month}$, Figure S1). We will verify it by field testing in our future work.

The coating property is critical for the hyperbranched polymer as the antifouling coating. Figure 3a and Table S1 show the adhesion strength and the pencil hardness grade using an epoxy panel as the substrate after immersion in ASW for different times. Before immersion, the adhesion strength of each coating is ~ 2.0 MPa because of the high content of acrylate segments in the polymer chain, which has a high polar interaction and hydrophobic interaction with the substrate.^{49,50} After immersion for 7 d in ASW, the adhesion strength of

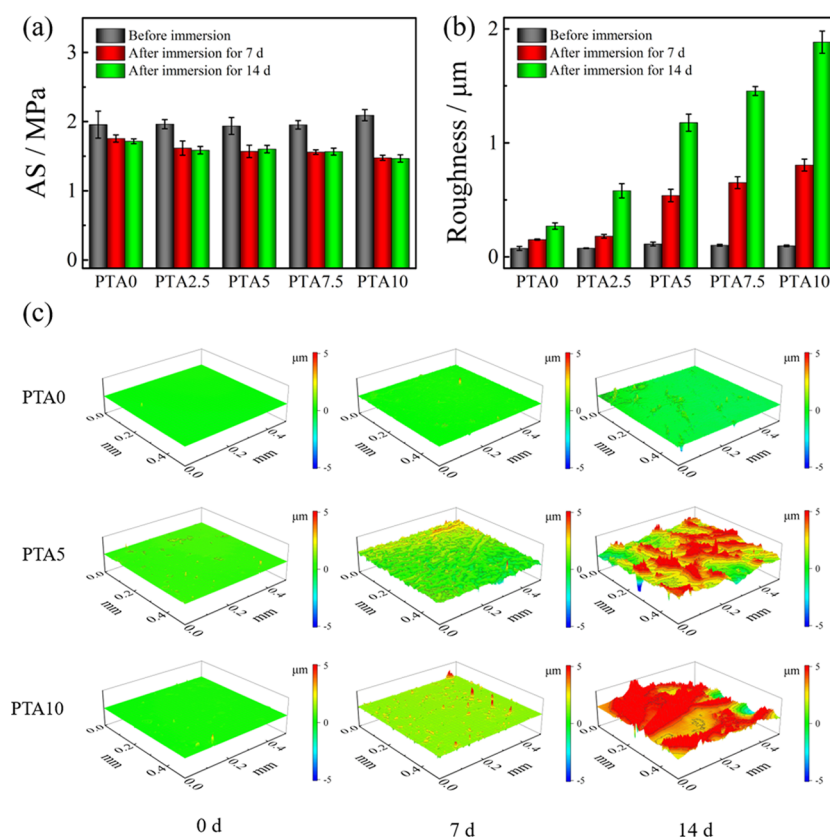


Figure 3. (a) Adhesion strength (AS) of PTAx to the epoxy substrate, (b) the roughness, and (c) the typical surface morphology of PTAx with different immersion times in ASW. Three parallel samples were measured to get the average value.

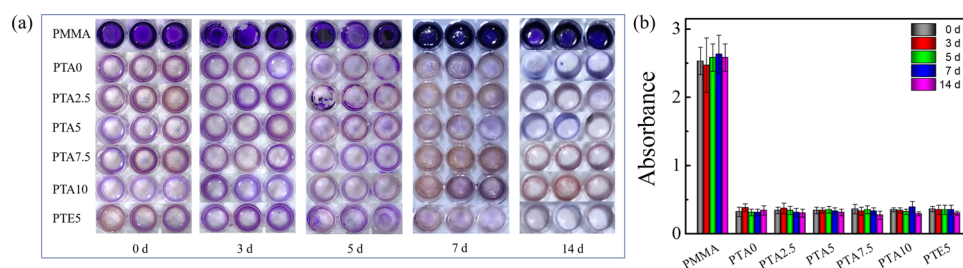


Figure 4. (a) *Pseudomonas* sp. biofilm grown on the polymer coatings with different pre-hydrolysis times stained with crystal violet; (b) absorbance of crystal violet from optical density tests. Three parallel samples were measured to get the average value.

PTAx slightly decreases and remains at ~ 1.5 MPa even after immersion for 14 d. This is probably because the hydrolysis occurs on the surface while the matrix remains hydrophobic. There is a slight difference among the PTAx coatings either before or after immersion in ASW due to the similar content of the TCB-TCPM in the polymer chain of PTAx (Table 1). Similarly, the hardness grade of all PTAx samples before immersion in ASW is *H*. For PTA0 and PTA2.5, the hardness grade remains at *H* during the immersion. For PTA5, PTA7.5, and PTA10, the hardness grade levels are at *B* after immersion. In other words, such polymeric coatings exhibit a relative stable coating property in marine environments.

Figure 3b,c shows the surface roughness and the typical surface morphology of PTAx cast-coated on an epoxy panel after immersion in ASW for different times at 25 °C. The surface roughness of all PTAx coatings increases with immersion time, indicating the occurrence of surface hydrolysis. For PTA0, the surface roughness slightly increases from ~ 0.10 to 0.25 μm after immersion for 14 d. As discussed

above, due to the absence of cleavable branched units in the PTA0 polymer chain, the degradation of PTA0 only relies on the hydrolysis of TCB-TCPM groups. Therefore, the degradation is so slow that the surface has slight changes in morphology. However, for PTA2.5, PTA5, PTA7.5, and PTA10, the surface roughness after immersion for 14 d is ~ 0.58 , ~ 1.18 , ~ 1.46 , and ~ 1.89 μm , respectively. This indicates that the changes of surface roughness increase with the contents of the cleavable branched units. The roughness has a slight effect on the biodegradation behavior since the initial roughness of PTAx is similar but they exhibit different degradation rates (Figure 2). Actually, the degradation rates depend on the contents of the cleavable branched units and a larger degradation rate leads to a rougher surface.

The antifouling ability of PTAx is evaluated using *Pseudomonas* sp. Nov.776 as a model biofouling since it is a typical marine bacterium isolated from subtidal biofilms in natural marine environments.⁵¹ We have monitored the growth curve of *Pseudomonas* sp. in an MB2216 broth (Figure S3).

The bacteria are in the mid-log phase at 12 h and are still in the stationary phase at 48 h. For the evaluation, the crystal violet is used to stain the biofilm on the polymer coatings on a 24-well microplate and the absorbance value is proportional to the amount of biofilm stained, where the substrates have a slight influence on the surface property of the coating. Figure 4 shows the *Pseudomonas* sp. biofilm grown on the polymer coatings with different prehydrolysis times stained with crystal violet and the absorbance of crystal violet, as determined from an optical density test. Before prehydrolysis, less biofilm is grown on the PTAx and PTE5 surface than on the PMMA control, indicating the excellent antifouling ability of the polymers. The excellent antifouling ability of PTAx and PTE5 before hydrolysis can be attributed to contact-kill mechanism because of the high killing efficiency of the TCPM groups. After prehydrolysis in ASW for 3 d, a significant amount of biofilm (absorbance: ~ 2.5) forms on the PMMA surface while a small amount of biofilm (absorbance: ~ 0.4) forms on the PTAx and PTE5 surface. This indicates that the polymers with TCB-TCPM groups still exhibit excellent antifouling ability even after immersion in ASW. As the prehydrolysis time increases to 14 d, the good antibiofilm ability (absorbance: ~ 0.4) of the PTAx and PTE5 surfaces is maintained. This is probably because of the generation of a zwitterionic surface of PTAx after hydrolysis, which resists the adsorption of bacteria. Since the anti-biofilm ability mainly relies on the functional group (TCB-TCPM) and all PTAx samples have a similar content of TCB-TCPM groups, the anti-biofilm formation performance of PTAx is almost the same. We also tested the anti-biofilm ability of PTAS during immersion in MB2216-containing bacteria with different times (Figure S4). A great amount of biofilm can be observed on the PMMA control surface with a high absorbance of ~ 2.6 during immersion, indicating that PMMA control is susceptible to the biofilm formation. However, for PTAS, the grown biofilm remains at a low level with an absorbance of ~ 0.4 during immersion, indicating the persistent anti-biofilm ability of PTAS in the bacterial suspension. We also conducted a plate counting assay to determine the number of surface-attached bacteria on the surface of PTAx during immersion in ASW-containing bacteria with different times (Figure S5). Similarly, almost no bacterial colonies can be observed on the plate of PTAS, indicating that there are no live bacteria attached onto the PTAS surface during immersion in ASW. In other words, the antifouling ability of the polymers is retained during immersion in ASW, probably due to the synergetic effect of "kill-resist-release".

To elucidate the underlying antifouling mechanism of PTAx during immersion in ASW, we examined the hydrolysis-induced zwitterions and their effect on the coating wettability and protein resistance. The samples were cast-coated on an epoxy panel for the CA, ATR-IR, and XPS tests. Figure 5a shows the ATR-IR spectra of PTAS before and after 1 h of hydrolysis under an accelerated condition in 0.1 M NaOH solution. The peaks at 3387 and 1576 cm^{-1} in the ATR-IR spectrum of PTAS after hydrolysis are attributed to the hydroxyl from hydrated zwitterions and carboxyls, respectively, indicating the generation of a zwitterionic surface. We also conducted XPS to demonstrate the coating chemistry of PTAS after hydrolysis (Figure 5b and Table 2). For the inner layer PTAS, the single peak at ~ 400.0 eV is attributed to N in the polymer. For the outermost layer of PTAS, the new peak at ~ 401.5 eV attributed to N^+ confirms the generation of a zwitterion. Note that the Cl element only exists in the TCPM

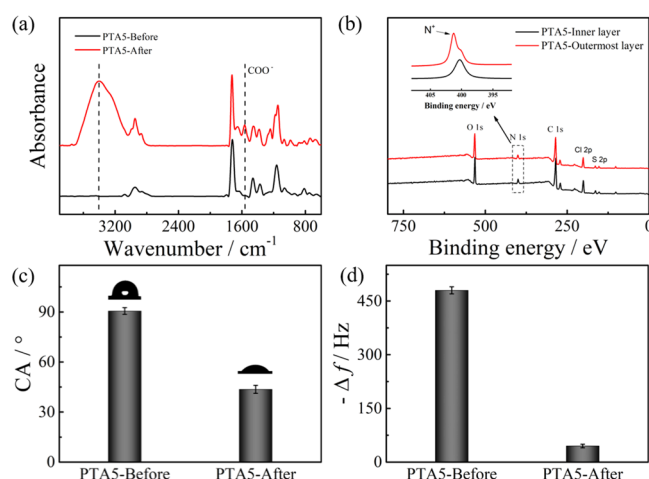


Figure 5. (a) ATR-IR spectra, (b) XPS spectra of the inner layer and the outermost layer of PTAS after hydrolysis in 0.1 M NaOH for 1 h, (c) water contact angle (CA), and (d) final frequency shift (Δf) in QCM-D measurements regarding protein adsorption. Five parallel samples were measured to get the average value of CA. Three parallel samples were measured to get the average value of Δf .

Table 2. Surface Elemental Composition of the Inner Layer and the Outermost Layer of PTAS after Hydrolysis in 0.1 M NaOH for 1 h Determined from XPS

sample	surface elemental concentration (atomic %)					Cl/C ratio
	C	N	O	Cl	S	
PTAS-inner layer	55.32	5.76	22.00	12.66	4.26	0.23
PTAS-outermost layer	76.26	3.01	17.26	2.60	0.87	0.03

groups and the lower ratio of Cl/C of the outermost layer than that of the inner layer indicates the release of the TCPM groups after hydrolysis. Figure 5c shows the water contact angle (CA) of PTAS before and after hydrolysis. The initial CA of PTAS is $\sim 90^\circ$, indicating the hydrophobic surface of PTAS. This is because the zwitterions are protected by TCPM groups and there are no hydrophilic groups in the polymer matrix. After hydrolysis, the CA is $\sim 45^\circ$, further indicating that the surface becomes hydrophilic owing to the generation of zwitterionic groups arising from the hydrolysis of TCB-TCPM. Note that the hydrolysis only occurs on the surface. The hydrolysis has a slight effect on its hydrophobic matrix. We also measured the surface energy of PTAS before and after hydrolysis (Figure S6). The higher surface energy of the hydrolyzed surface further indicates the generation of a zwitterionic surface. Protein resistance is an important characteristic of "defending" antifouling materials. The initial process of biofouling involves the adsorption of protein on the surface, thereby providing a platform for bacterial colonization and biofilm formation.⁵² Therefore, protein resistance is critical for marine antifouling. We evaluated the adsorption of proteins onto the PTAS surface via QCM-D (Figure 5d). The substrate is not an epoxy panel but an AT-cut quartz crystal since the polymer must be spin-coated on its surface for the QCM-D test. The in situ QCM-D curves are shown in Figure S7. Fibrinogen can strongly adsorb onto immersed surfaces and is therefore taken as the model protein. The shift in frequency (Δf) is related to the mass change of the thin layer. The final Δf of PTAS before hydrolysis is about -460 Hz, indicating

that the protein strongly adsorbs onto the non-hydrolyzed surface. In other words, PTAS before hydrolysis performs no “defending” function, which further indicates that the antibiofilm ability of PTAS before hydrolysis (Figure 4) arises from its “attacking” function. The Δf of PTAS after hydrolysis is considerably lower than that before hydrolysis, indicating that significantly less protein adsorption occurs on the surface. As reported before, higher surface roughness would lead to a higher protein adsorption.⁵³ However, the hydrolyzed surface in the study has a lower protein adsorption even the hydrolyzed surface is rougher than the initial surface. Thus, the surface chemistry (zwitterion) has a larger effect on protein adsorption behavior than the surface roughness. In other words, PTAS after hydrolysis performs a “defending” function owing to the generation of a zwitterionic surface.

To further confirm the antifouling mechanism of PTAS coatings, we evaluated the antibacterial efficiency of PTAS before and after hydrolysis via live/dead assays, where PTAS was used as the typical sample since all the samples have similar contents of TCB-TCPM groups. Figure 6a shows the

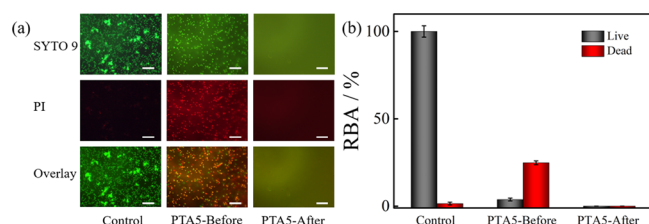


Figure 6. (a) Fluorescence photographs and (b) relative bacterial adsorption (RBA) of *Pseudomonas* sp. adhered on the polymer surfaces—green color indicates all the adsorbed bacteria (SYTO 9) and red color indicates dead bacteria (PI). Scale bar: 50 μm . Three parallel samples were measured to get the average value.

fluorescence microscopy images of PTAS surfaces after a 5 h immersion period in an ASW suspension of marine bacteria *Pseudomonas* sp. Figure 6b shows the relative bacterial adsorption (RBA) obtained by the CFU of coatings compared with CFU of the control sample, which can reflect the direct difference of the antibacterial ability of the coating compared with the same control. All the adsorbed bacteria can be stained green by SYTO 9, and the dead bacteria can be stained red by PI. For PMMA control, numerous bacteria (green, RBA: ~100%) and a few dead bacteria (red, RBA: ~1%) are observed on the surface, indicating that PMMA is susceptible to the bacterial adhesion. Before hydrolysis, a large group of dead bacteria (RBA: ~25%) and a small group of live bacteria (RBA: ~3%) are observed on the surface, indicating that PTAS is effective in “attacking” the bacteria. The PTAS surface after hydrolysis is almost bacteria-free (RBA: <1%) indicating that the hydrolysis-generating zwitterionic surface is effective in “defending” the adhesion of bacterial cells. Note that the coating surface is hydrophilic and rough after hydrolysis, the dye can easily permeate into the coating surface and hardly be washed away. Thus, the background of PTAS-After is stained. Moreover, we observed the morphology of surface-attached bacteria on PTAS before and after hydrolysis using SEM (Figure S8). The cell wall membranes of surface-attached bacteria on PMMA control look to be intact, smooth, and unspoiled, whereas it is collapsed and distorted on PTAS before hydrolysis, indicating that the bacteria are killed on the PTAS surface through a contact–killing mechanism because of

the high killing efficiency of the TCPM groups. Almost no surface-attached bacteria on the PTAS surface after hydrolysis can be observed, indicating the excellent resistance ability of the generated zwitterionic surface after hydrolysis. In other words, PTAS exhibits good antifouling efficiency through the “attacking” and “defending” strategy.

CONCLUSIONS

In conclusion, a hydrophobic dual-functional monomer–tertiary carboxybetaine ester acrylate with the antifouling group *N*-(2,4,6-trichlorophenyl)maleimide (TCB-TCPM) is synthesized and then copolymerized with methacrylic anhydride via reversible addition-fragmentation chain transfer (RAFT) polymerization to yield a degradable hyperbranched polymer. Such a polymeric coating is characterized by a controllable degradation rate, which increases with the degrees of branching. Moreover, it exhibits excellent antifouling abilities against protein, marine bacteria *Pseudomonas* sp. during immersion in ASW via the synergistic effect of “attack” and “defense”. Namely, the hydrolysis of the TCB-TCPM group allows the antifouling group TCPM to leach out to kill bacteria and generates a zwitterionic surface to resist the adhesion of protein and bacteria. Particularly, the degradation of hyperbranched polymers makes the surface fragmenting, leading to a self-regenerating attacking and defending dynamic surface. The strategy “kill–resist–renew trinity” can provide a highly promising platform for antifouling coatings.

ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acsami.1c02273>.

Experimental section, changes of thickness for PTAS, release rate and cumulative release of TCPM from PTAS, pencil hardness of PTAS, growth curve of *Pseudomonas* sp., crystal violet assay and plate counting assay on PTAS, surface energy of PTAS, in situ QCM-D curves, and SEM images of the bacterial morphology (PDF)

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Notes

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